Bartosz Kruszewski

SEMINAR: LARGE LANGUAGE MODELS

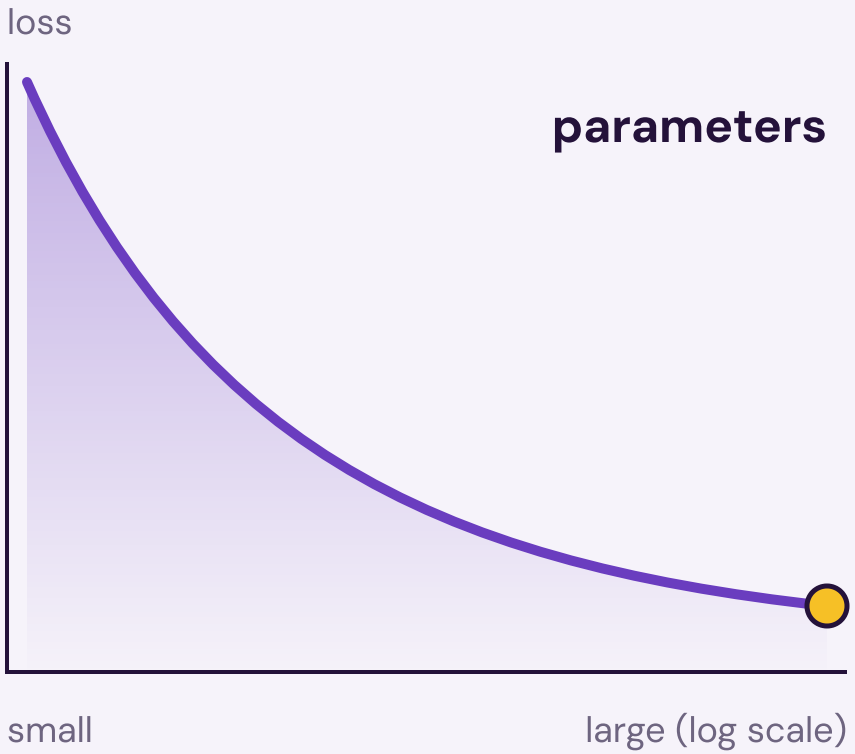
Textbooks Are All You Need

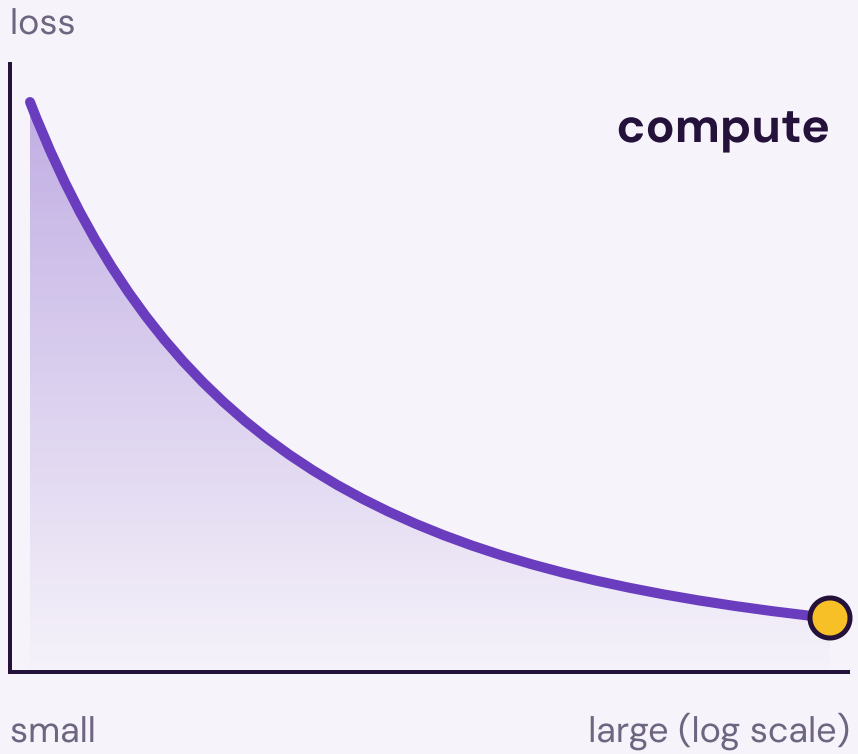
Gunasekar et al. · Microsoft Research · 2023

What if *better data* beats bigger models?

Scaling laws

more parameters, more compute → lower loss





Kaplan et al., 2020 · Hoffmann et al., 2022 (cited in the paper) · TinyStories: Eldan & Li, 2023

The domain: code

write a Python function from its docstring

```python
def incr_list(l: list):
    """Return list with elements
    incremented by 1."""
    return [x + 1 for x in l]
```

HumanEval

164 problems · unit tests

MBPP

mostly basic Python problems

pass@1

one try, must pass all tests

phi-1 at a glance

1.3B
parameters

7B
tokens of data

4 days
on 8 × A100

50.6%
HumanEval pass@1

HumanEval vs open & closed models

pass@1, self-reported · June 2023

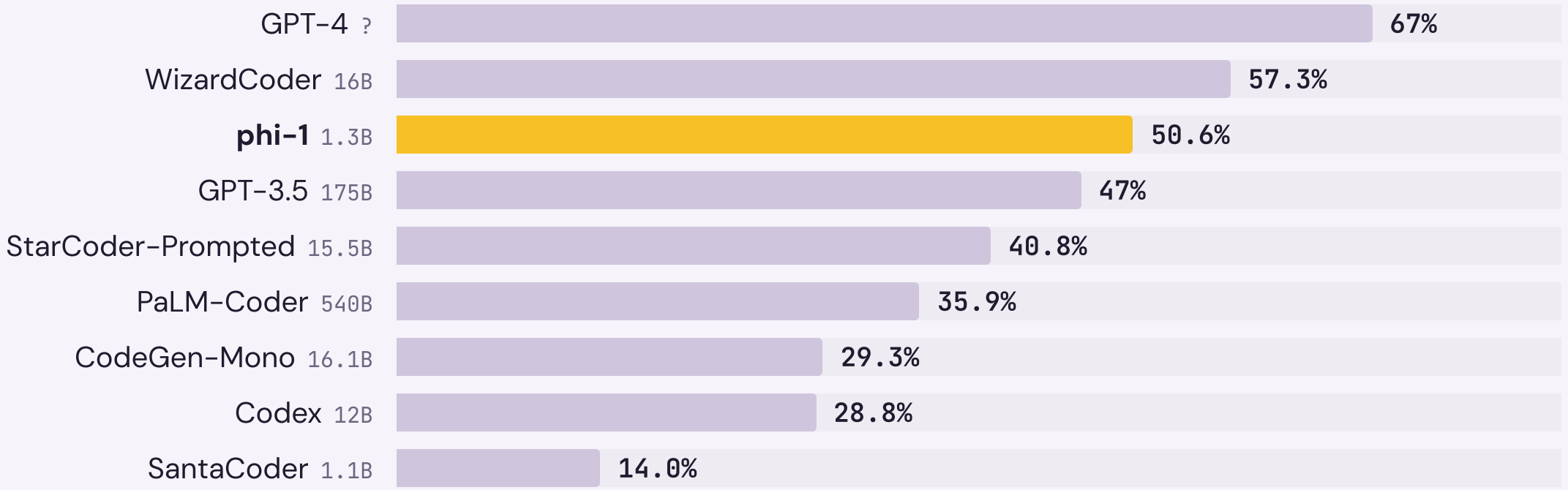


Table 1 of the paper

Size vs score

HumanEval pass@1 vs parameters

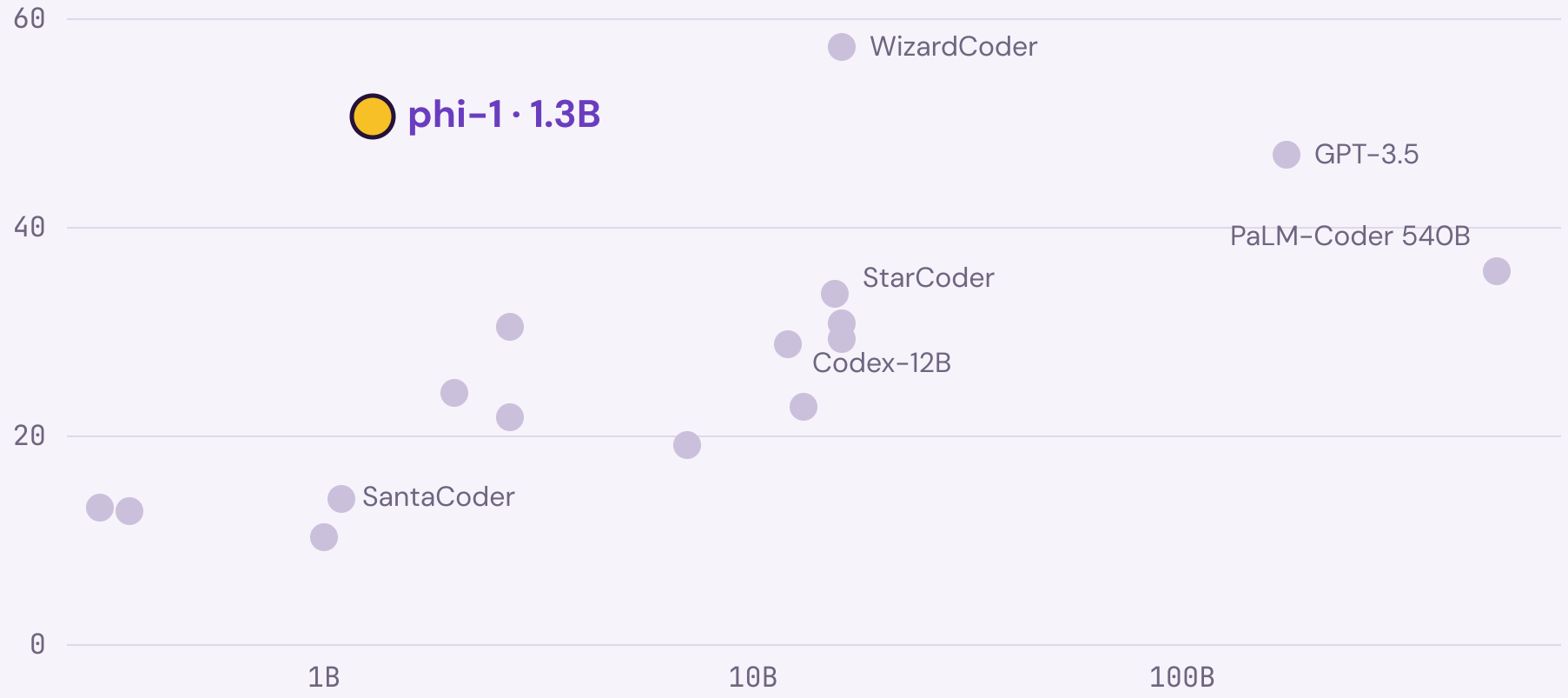


Table 1 · x axis log scale

What's wrong with web code?

The Stack · StackOverflow — huge, diverse, but not instructive

01

Not self-contained

depends on other files

02

Boilerplate

constants, configs, GUI setup

03

Buried logic

inside long, undocumented functions

04

Skewed topics

unbalanced coverage of skills

High vs low educational value

✓ HIGH EDUCATIONAL VALUE

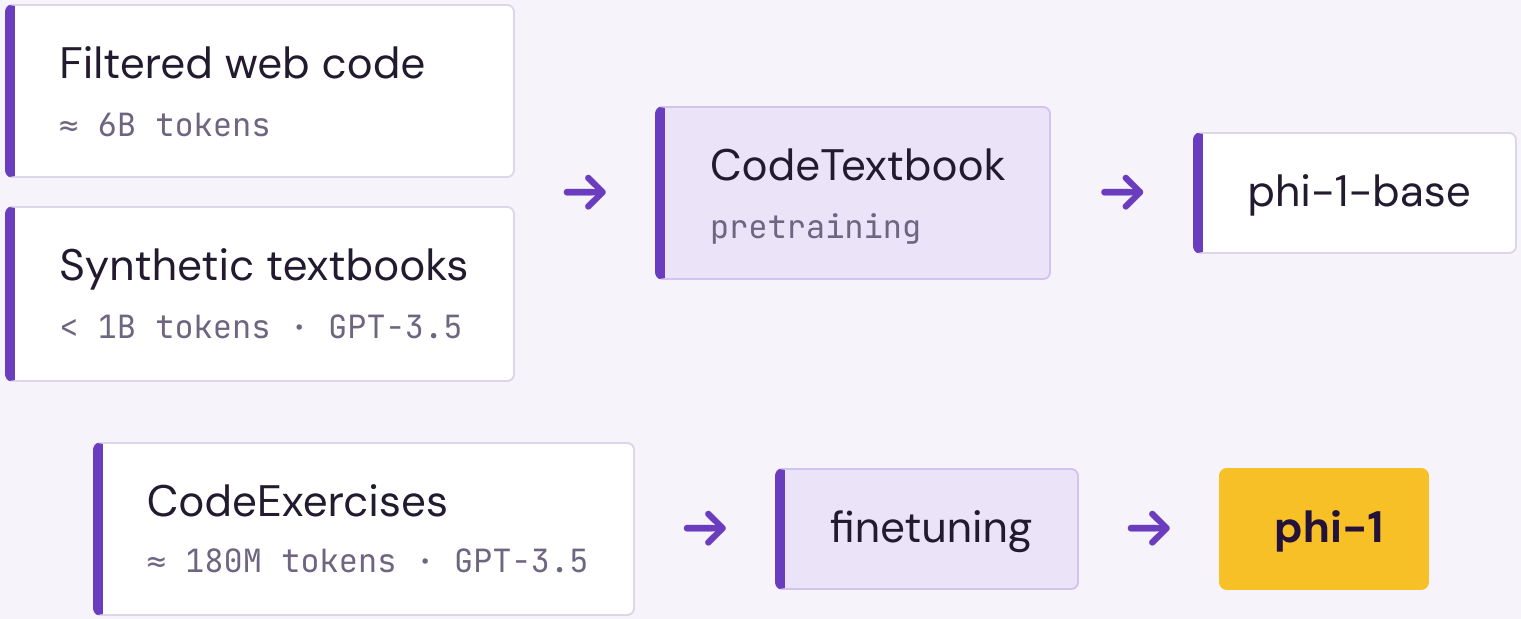
```
def cosine_dist(x, y):
    """Computes Cosine Distance."""
    x = F.normalize(x, dim=1)
    y = F.normalize(y, dim=1)
    dist = 2 - 2 * torch.mm(x, y.t())
    return dist
```

✗ LOW EDUCATIONAL VALUE

```
class Default(object):
    def __init__(self, vim):
        self._vim = vim
        self._bufnr = -1
        self._winid = -1
        self._winrestcmd = ''
        self._initialized = False
```

A good textbook is *clear*, *self-contained*, *instructive*, *balanced*

The data recipe



Step 1 · Filter the web

The Stack + SO
35M files · 35B tokens

→

GPT-4 labels
~100k samples

→

random forest
on CodeGen embeddings

→

≈ 6B tokens

prompt: *“determine its educational value for a student whose goal is to learn basic coding concepts”*

Filtering alone helps

350M model · HumanEval pass@1



Step 2 · Synthetic textbooks: diversity

GPT-3.5 · LLMs follow the most probable path



naive prompting



random topic + random audience

A synthetic textbook page

“A matrix is said to be singular if its determinant is zero… Now, let's explore these concepts through examples.”

```
import numpy as np

def is_singular(A):
    det = np.linalg.det(A)
    return det == 0

A = np.array([[1, 2], [2, 4]])
print(is_singular(A))  # True
```

Step 3 · CodeExercises

≈ 180M tokens · docstring → solution · diversity via function names

```
def valid_guessing_letters(word: str, guesses: List[str]) → List[str]:
    """Returns a list of valid guessing letters, which are letters
    that have not been guessed yet and are present in the word."""
    valid_letters = []
    for letter in word:
        if letter not in guesses and letter not in valid_letters:
            valid_letters.append(letter)
    return valid_letters
```

Model & training

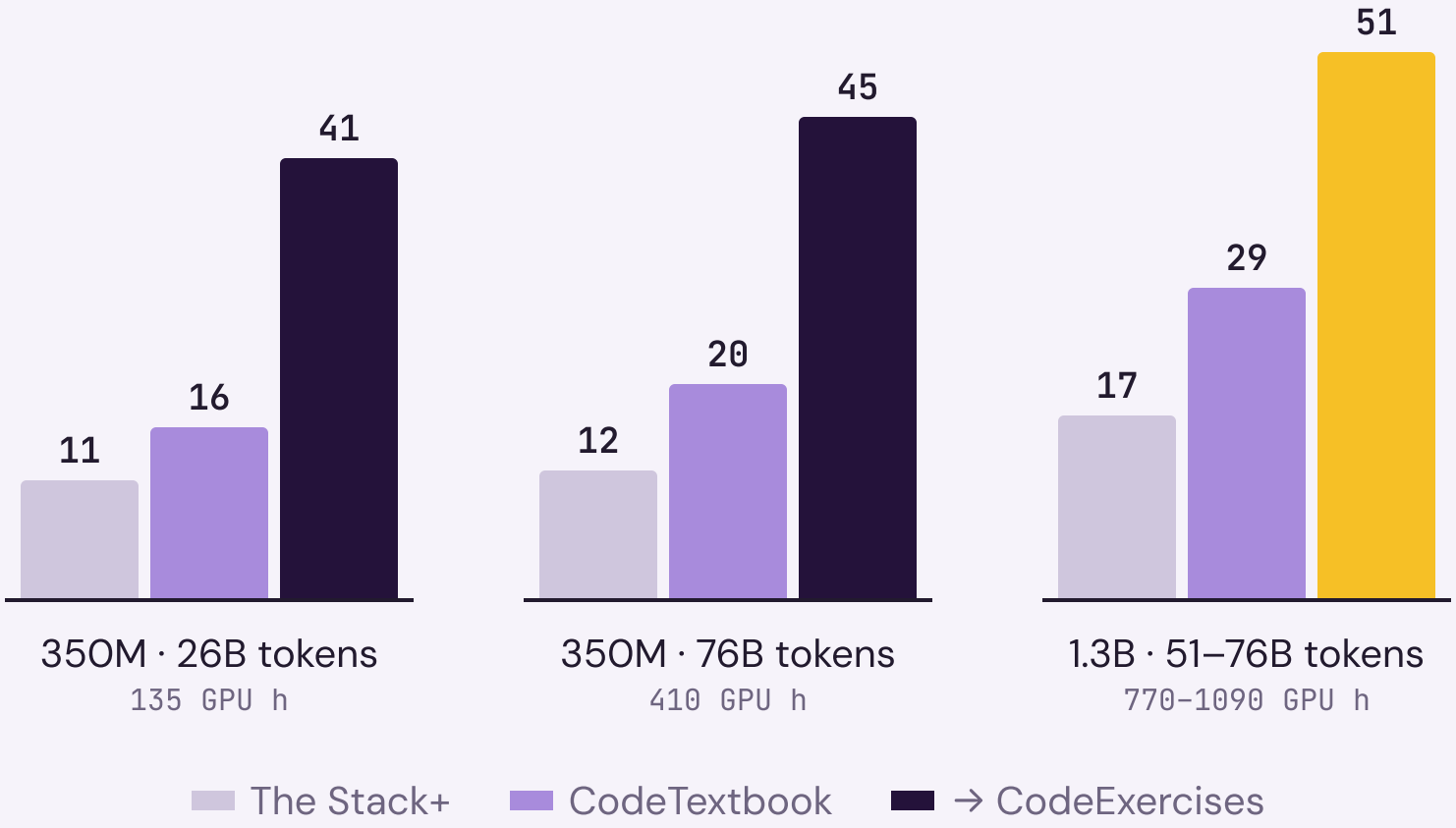
decoder-only transformer, deliberately plain

	PHI-1	PHI-1-SMALL
Parameters	1.3B	350M
Layers	24	20
Hidden / MLP dim	2048 / 8192	1024 / 4096
Heads × dim	32 × 64	16 × 64

FlashAttention | **parallel MHA + MLP** | **RoPE** | ✕ FIM | ✕ MQA

The key experiment

HumanEval pass@1 (%) · Figure 2.1



180M tokens of exercises:

29% → 51%

Emergent skills · understanding

finetuning on simple exercises unlocks skills not in the exercises

PHI-1

✓ correct

follows every rule

PHI-1-BASE

✗ rambles

endless `self.x = …`

PHI-1-SMALL

~ partial

gets the idea, wrong logic

prompt: Alice, Bob and Charles pick numbers… simulate 30 rounds

Emergent skills · libraries

PyGame & Tkinter — absent from the exercises

	KNOWS THE API	FOLLOWS THE LOGIC
phi-1	✓	✓
phi-1-base	✓	✗
phi-1-small	✗	✓

Memorization? · 50 new problems

separate team · designed to be unusual · graded 0–10 by GPT-4

```
def sort_concat_square_deduplicate(list1, list2, my_threshold):
    """Sorts each list, concatenates them, squares the entries
    at even indices, filters out entries smaller than
    my_threshold and then removes duplicates."""
```

Same ranking on new problems

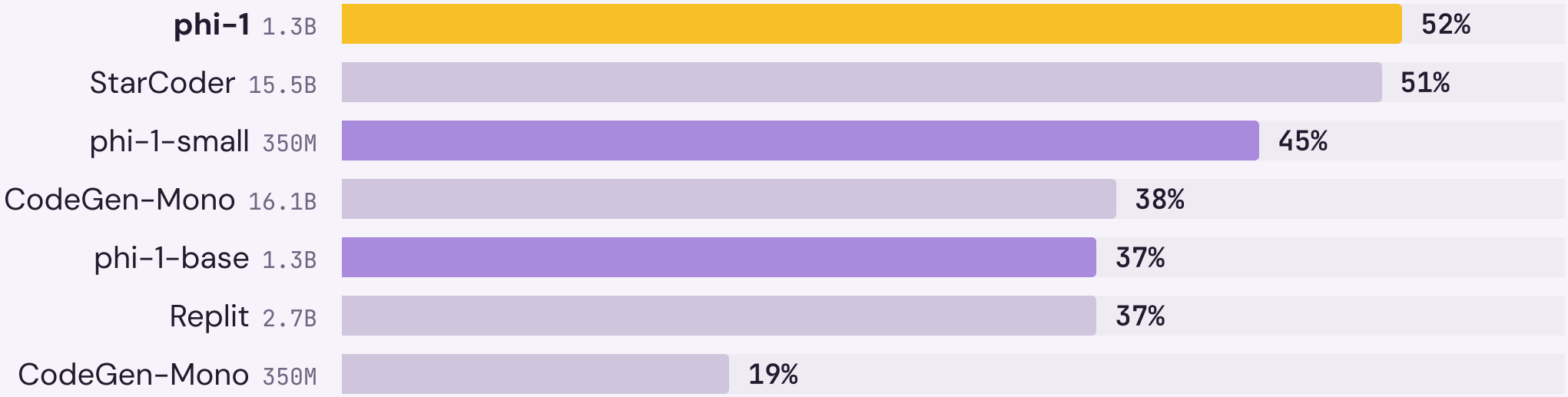


Table 2 · LLM-graded understanding score on 50 unconventional problems

n-gram overlap

13	4	0
-gram window	HumanEval hits	real matches

all 4 were false positives — shared phrase “frequency of an integer…”

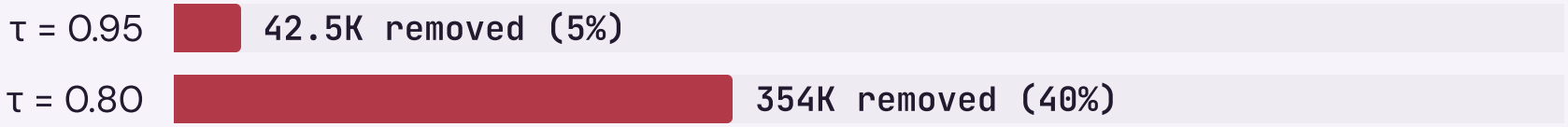
Prune anything similar

embedding distance
CodeGen-350M

+

AST edit distance
match rate τ

→ remove similar → **retrain phi-1**



Still beats StarCoder

HumanEval pass@1 after retraining on pruned data

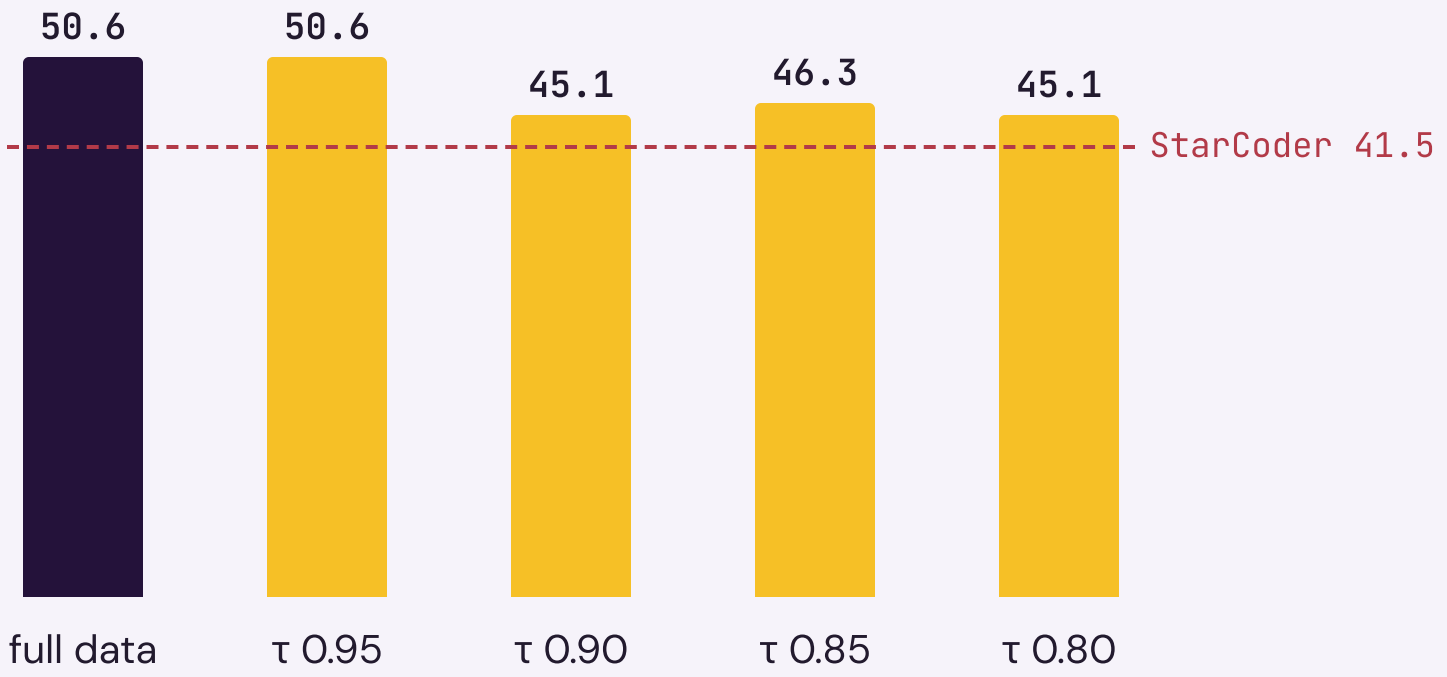


Table 3

Limitations

SCOPE

Python only

narrow APIs, no complex apps

PROMPTS

long prompts fail

3 layers ✓ · 4 layers ✗ · + `import torch` ✓

LANGUAGE

typos & wording

“x stays unchanged” → `x = 0`

REASONING

counting & layout

extra text field, wrong buttons

Open problems they name

01

Coverage

all concepts, balanced

02

Diversity

no repetition

03

Measurement

how to quantify diversity?

04

Ethics

LLMs curating data for LLMs

paper in one line: 10× smaller model · 100× less data · near-SOTA on code

BEYOND THE PAPER

What happened next

the Phi family and its impact · 2023 → 2026

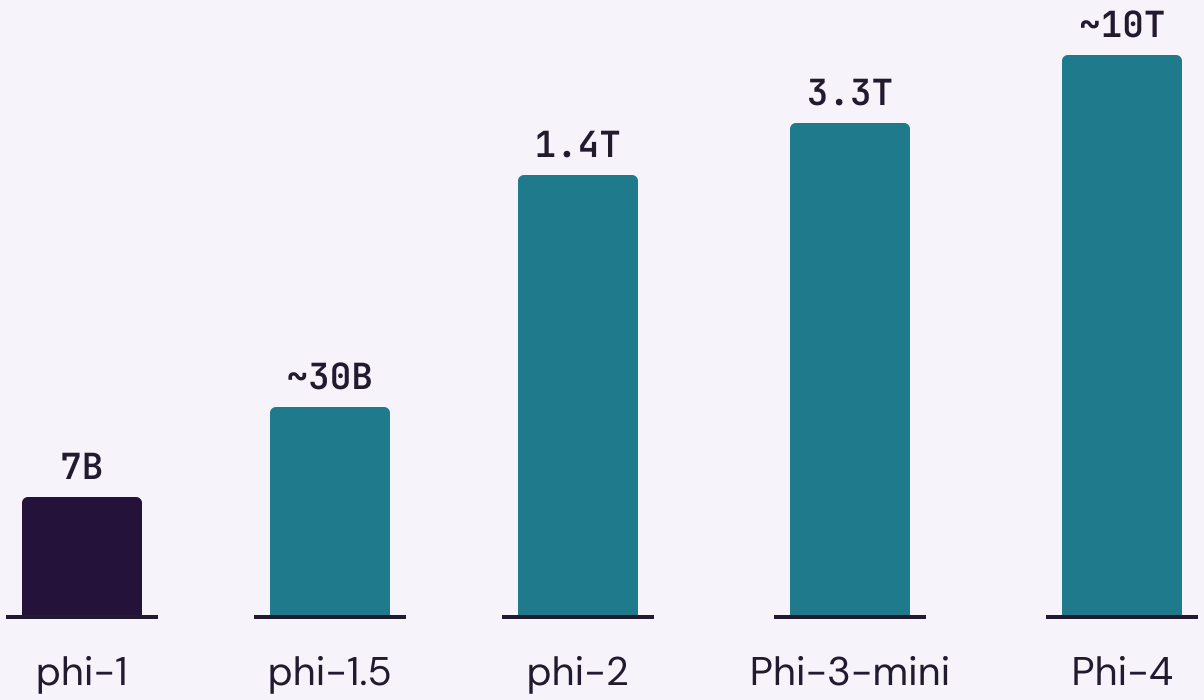
The Phi family

Jun 2023	Sep 2023	Dec 2023	Apr 2024	Dec 2024	Apr 2025	Mar 2026
phi-1	**phi-1.5**	**phi-2**	**Phi-3**	**Phi-4**	**Phi-4-reasoning**	**Phi-4-r-vision**
1.3B · code this paper	1.3B · common sense	2.7B · general	3.8B–14B on a phone	14B · mostly synthetic	14B · thinks	15B · knows when to think

Microsoft tech reports, Azure blog, VentureBeat

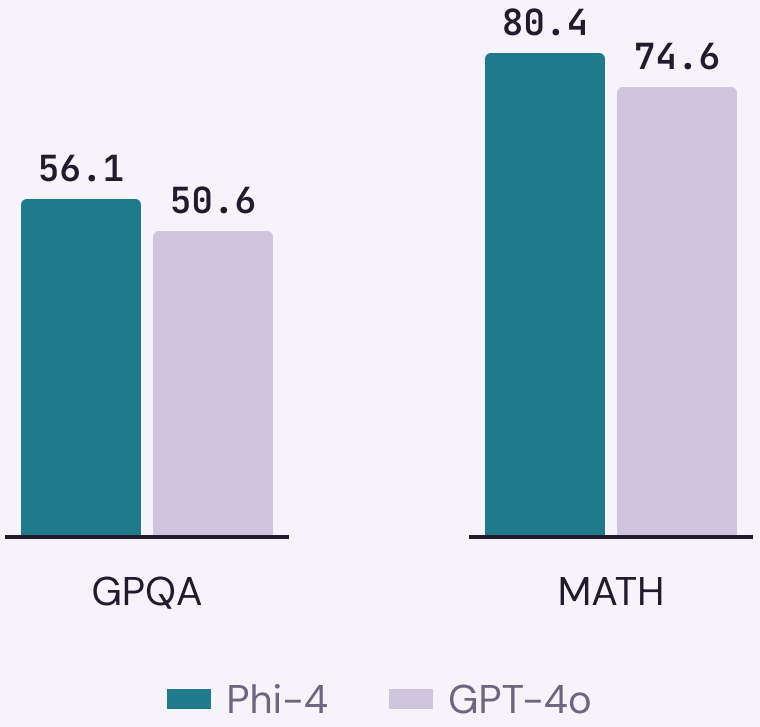
Small model ≠ small data anymore

unique training tokens · log scale



Phi-4 · student beats teacher

14B · ~400B synthetic tokens · vs its data generator GPT-4o



Phi today · October 2026

MODELS

No Phi-5 yet

Phi-4 line keeps specializing

WHERE

Windows, Azure, HF

Phi Silica on-device · MIT weights

PEOPLE

Bubeck → OpenAI

Oct 2024 · most co-authors stayed

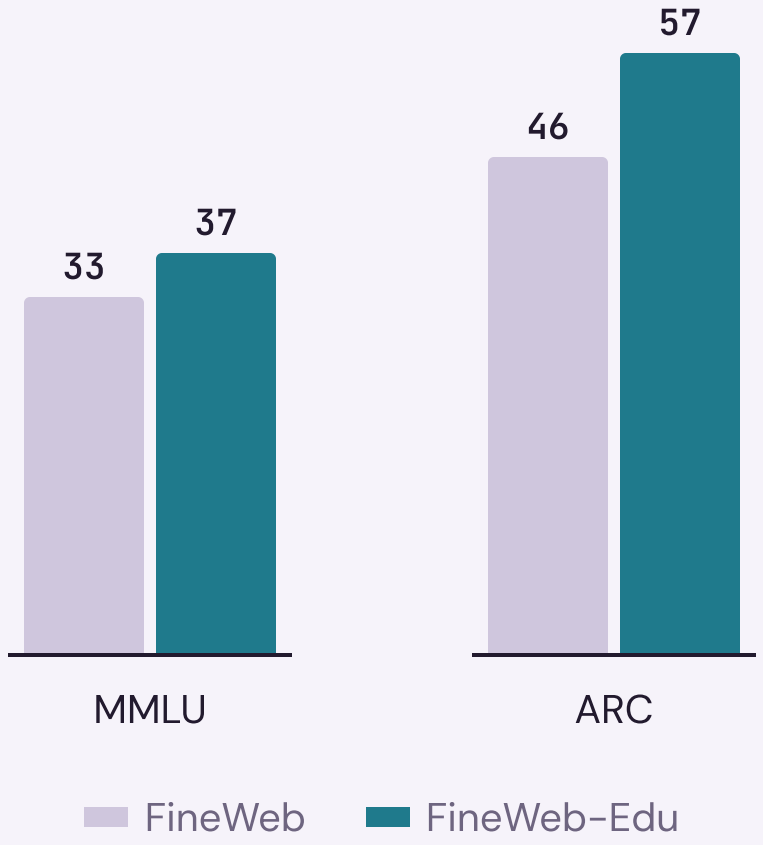
MICROSOFT

Also MAI models

in-house large models next to Phi

Impact · “educational value” filters

FineWeb-Edu (Hugging Face, 2024) · Llama-3-70B labels → classifier → keep score ≥ 3/5



Penedo et al., 2024 · arXiv 2406.17557

Impact · synthetic data goes mainstream

OPEN TEXTBOOKS

Cosmopedia → SmolLM

~25B tokens by Mixtral · data + code released

REWRITE THE WEB

rephrase as QA, tutorials

WRAP · Nemotron-CC · BeyondWeb

DISTILLATION

teacher writes the textbook

GPT-4o → Phi-4 · R1 → small reasoners

Critiques

SEP 2023 · PARODY

“Pretraining on the Test Set Is All You Need”

1M-param phi-CTNL “beats everything”

COMMUNITY

benchmarks » vibes?

strong scores, weaker in messy real use

NATURE 2024

model collapse

recursive training on own outputs degrades models

Did the paper's claims hold up?

CLAIM IN THE PAPER	TODAY
data quality bends scaling laws	✓ widely accepted
LLM-generated training data works	✓ standard practice
LLM-labelled quality filters	✓ FineWeb-Edu, Llama 3
tiny data is enough	✕ Phi now uses trillions
benchmarks reflect real skill	~ still debated

Takeaways

01

Data quality is a scaling axis

02

A small finetune can unlock skills

03

Check contamination seriously

Thank you

Questions?

Sources

Gunasekar et al., 2023
Textbooks Are All You Need · arXiv:2306.11644

Kaplan et al., 2020
Scaling Laws for Neural LMs · arXiv:2001.08361

WEB **Abdin et al., 2024**
Phi-3 Technical Report · arXiv:2404.14219

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Phi-4-reasoning · arXiv:2504.21318

WEB **Hugging Face, 2024**
Cosmopedia blog · huggingface.co/blog/cosmopedia

WEB **Shumailov et al., 2024**
AI models collapse… · Nature 631

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Chen et al., 2021
Evaluating LLMs Trained on Code · arXiv:2107.03374

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Phi-4 Technical Report · arXiv:2412.08905

WEB **Penedo et al., 2024**
The FineWeb Datasets · arXiv:2406.17557

WEB **Schaeffer, 2023**
Pretraining on the Test Set Is All You Need · arXiv:2309.08632

WEB **VentureBeat, Mar 2026**
Phi-4-reasoning-vision-15B